

ANINDYA MONDAL

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PhD Candidate (AI) — Research focus: multimodal reasoning, VLMs, agentic workflows.

EDUCATION

University of Surrey

Guildford, UK

PhD Candidate in Artificial Intelligence

Oct 2022 – Present

- Surrey Institute for People-Centred AI, CVSSP. Research on multimodal learning, vision-language integration, generative models and agent-guided generation.

Jadavpur University

Kolkata, India

B.E. (Hons.) in Electronics & Telecommunication Engineering

Aug 2018 – Jun 2022

- GPA: 8.79/10

RESEARCH FOCUS (SELECTED)

- Vision-language integration, agent-guided generation, tool-augmented reasoning, multimodal foundation models, planning and decision-making for autonomous agents, and multi-agent collaboration.
- Applied research on text-to-image/video generation, high-instance scene creation, and object counting.

SELECTED RELEVANT PROJECTS

CountLoop (Project)

2024 – 2025

VLM agent-guided high-instance image generation

University of Surrey

- Designed and implemented **CountLoop**, a vision-language-model (VLM) agent that iteratively guides a diffusion generator to control instance counts and layout, enabling precise high-instance scene synthesis without retraining the generator.
- Work involved agent orchestration, prompt-engineering for multimodal models, tool integration for counting and layout verification, and evaluation on count-accuracy and aesthetic alignment.

Video Generation Acceleration & Control Pipeline

2025 – 2026

Optimising diffusion-based video generation

University of Surrey

- Developing a lightweight pipeline to accelerate diffusion-based video generation while preserving temporal coherence and motion fidelity.
- Designed compact conditioning and scheduling strategies that reduce computational load and improve responsiveness in long-sequence generation.
- Work includes building verification and correction loops that stabilise motion and ensure consistency across frames without exposing proprietary architectural details.

OmniCount (AAAI 2025)

2025

Class-agnostic object counting

University of Surrey

- Built a multi-label counting model that leverages semantic and geometric priors for robust counting in crowded/high-instance scenes; focused on generalization across categories and integration with vision-language feedback loops.

MSQNet (ICCVW 2023)

2023

Multimodal Transformer for Action Recognition

University of Surrey

- Proposed MSQNet, a transformer-based multimodal action-recognition model using multi-modal query fusion; achieved significant precision gains and contributed to understanding cross-modal reasoning mechanisms relevant to agent perception.

RESEARCH & TECHNICAL EXPERIENCE

University of Surrey

Guildford, UK

Doctoral Researcher

Oct 2022 – Present

- Led research projects combining foundation models and agent-like orchestration for improving controllability in generative models.
- Developed evaluation protocols and automated tool-use checks (count verification, layout matching) to close the loop between model outputs and agent decisions.

Indian Institute of Science

Bengaluru, India

Research Intern

May 2022 – Aug 2022

- Devised a source-free domain-adaptation framework for image classification to improve robustness under domain shift.

Jadavpur University

Kolkata, India

Undergraduate Research Assistant

Oct 2020 – May 2022

- Worked on graph-based learning and event-based vision for moving-object detection; implemented spectral methods and graph neural network pipelines.

PUBLICATIONS (SELECTED)

CountLoop: Iterative Agent Guided High Instance Image Generation.

A. Mondal, A. Banerjee, S. Nag, J. Lladós, X. Zhu, A. Dutta. Preprint. [Project]

OmniCount: Multi-label Object Counting with Semantic–Geometric Priors.

A. Mondal, S. Nag, X. Zhu, A. Dutta. AAAI 2025.

Actor-Agnostic Multi-label Action Recognition with Multi-modal Query.

A. Mondal, S. Nag, J.M. Prada, X. Zhu, A. Dutta. ICCVW 2023.

TECHNICAL SKILLS

- **Languages:** Python (expert), MATLAB (expert), C (advanced).
- **Frameworks:** PyTorch (expert), TensorFlow (familiar), scikit-learn, NumPy, SciPy, Pandas.
- **Agent/LLM tooling:** Familiar with building tool-augmented workflows using VLMs/LLMs, prompt engineering, evaluation loops, and programmatic tool invocation for verification and planning.
- **Reinforcement learning (relevant):** Working knowledge of RL fundamentals (policy gradients, value-based methods) and experience prototyping RL components and reward structures in PyTorch for research experiments.
- **Other tools:** Git, Docker, Jupyter Notebook, $\LaTeX$.

AWARDS AND FUNDING

AAAI 2025 Conference & Travel Grant, Philadelphia, USA.

ICCV 2023 Conference Grant, Paris, France.

Postgraduate Studentship, University of Surrey (2022).

Uplink Research Internship Award, ACM SIGKDD India Chapter (2022).

PROFESSIONAL EXPERIENCE & SERVICE

University of Surrey

Guildford, UK

Teaching

2023 – 2025

- Demonstrator: Applied Machine Learning (EEEM068); Advanced Topics in Computer Vision & Deep Learning (EEEM071).
- Teaching Fellow: Applied Machine Learning for University of Surrey MSc People-Centred Artificial Intelligence (edX).

Reviewer

Conferences & Journals

2022 – 2025

- Reviewer for CVPR, ICCV, ECCV, NeurIPS, ICASSP, BMVC, IEEE TSP, TSIPN, ISBI.